

1. (15 pts) Let $A = \{\dots, -8, -4, 0, 4, 8, 12, \dots\}$, $B = \{x \in \mathbb{Z} : -2 < x \leq 1\}$, $C = \{1, 2\}$.

(a) Write the set A using set builder notation.

(b) Find $C \times B$.

(c) Is $A \subseteq \mathbb{N}$? Why or why not?

2. (10 pts) Construct the truth table for $(\sim Q \wedge P) \implies (P \vee Q)$. Insert any needed columns and be sure to label them.

P	Q	

3. (10 pts) Write the negation of each of the following statements. You may use words, symbols, or a combination of both. **(Do not prove these statements.)**

(a) If a is even or $5 \nmid b$ then $2a + 4b \leq 25$ and $3 \mid (a - b)$.

(b) $\forall x, y \in \mathbb{Z}, \exists z \in \mathbb{Z}$ such that $((y \mid z) \vee (x + z \text{ is odd}))$.

4. (10 pts) Prove the following statement:

Suppose $x, y \in \mathbb{Z}$. If $10 \nmid 3xy$ then x is odd or $5 \nmid y$.

5. (15 points) Consider the following new definition:

Definition: An integer x is said to be **fourven** if $x = 4a + 2$ for some $a \in \mathbb{Z}$

(a) Is 10 fourven? Why or why not?

(b) Is 12 fourven? Why or why not?

(c) Let $x, y \in \mathbb{Z}$. Prove that if x or y are even then $2xy + 6$ is fourven.

6. (10 pts) Prove the following statement:

If R is a rational number and $R - 3I$ is an irrational number, then I is irrational.

7. (10 pts) Let $a \in \mathbb{Z}$. Prove that $5a^2 + 7$ is even if and only if a is odd.